

CHAPTER – XII

APPENDIX

12.1 – Source Code

app.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def home():
    return render_template('index.html')

@app.route('/about')
def about():
    return render_template('about.html')

@app.route('/dashboard01')
def dashboard01():
    return render_template('dashboard01.html')

@app.route('/dashboard02')
def dashboard02():
    return render_template('dashboard02.html')

@app.route('/story')
def story():
    return render_template('story.html')

if __name__ == '__main__':
    app.run(debug=True)
```

base.html

```
<!DOCTYPE html>
<html>
<head>

<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
```

Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

```
<title>EV Analytics</title>

<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap
.min.css" rel="stylesheet">

<link rel="stylesheet"
href="{{ url_for('static', filename='css/style.css') }}">

</head>

<body>

<nav class="navbar navbar-expand-lg navbar-dark">

<div class="container">

<a class="navbar-brand"
href="{{ url_for('home') }}">
EV Analytics
</a>

<div class="navbar-nav ms-auto">

<a class="nav-link"
href="{{ url_for('home') }}">Home</a>

<a class="nav-link"
href="{{ url_for('about') }}">About</a>

<a class="nav-link"
href="{{ url_for('dashboard01') }}">Dashboard 1</a>

<a class="nav-link"
href="{{ url_for('dashboard02') }}">Dashboard 2</a>

<a class="nav-link"
href="{{ url_for('story') }}">Story</a>

</div>

</div>

</nav>
```

Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

index.htm

```
{% block content %}
{% endblock %}

</body>
</html>

{% extends 'base.html' %}

{% block content %}

<style>

    .hero{
    min-height:100vh;

    background-image:
    linear-gradient(
    rgba(0,0,0,0.20),
    rgba(0,0,0,0.35)
    ),
url('{{ url_for("static", filename="images/ev-bg.jpg") }}');

    background-size:cover;
    background-position:center;
    background-repeat:no-repeat;

    display:flex;
    align-items:center;
    justify-content:center;

    text-align:center;
    }

    .hero-content{
    max-width:900px;
    padding:30px;
    }

    .hero h1{
    font-size:4rem;
    font-weight:700;
    color:white;

    text-shadow:
    2px 2px 15px rgba(0,0,0,0.7);
    }
```



Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

```
}

.hero p{
font-size:1.3rem;
color:#f1f5f9;

text-shadow:
1px 1px 10px rgba(0,0,0,0.7);
}

.btn-hero{
background:linear-gradient(
135deg,
#2563EB,
#60A5FA
);

border:none;
color:white;

padding:14px 35px;

border-radius:50px;

font-weight:600;

transition:.3s;
}
```

```
.btn-hero:hover{
transform:translateY(-3px);

box-shadow:
0 0 20px rgba(96,165,250,.5);

color:white;
}
```

```
.kpi-section{
background:#050816;
padding:70px 0;
}
```

```
.kpi-card{
background:#0F172A;
```

```
border:1px solid #1E293B;

border-radius:20px;

padding:30px;

text-align:center;

color:white;

transition:.3s;
}

.kpi-card:hover{
transform:translateY(-5px);

box-shadow:
0 0 25px rgba(59,130,246,.25);
}

.kpi-card h2{
color:#60A5FA;
font-weight:700;
}

</style>
```

```
<!-- Hero Section -->
```

```
<section class="hero">
```

```
<div class="hero-content">
```

```
<h1>
```

Electric Vehicle Charge & Range Analysis

```
</h1>
```

```
<p class="lead mt-4">
```

Advanced Visualization Platform for EV Charging,
Battery Performance, Energy Consumption
and Driving Range Analytics

```
</p>
```

```
<a href="{{ url_for('dashboard01') }}"
```

Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

class="btn btn-hero btn-lg mt-3">

Explore Dashboard

</div>

</section>

<!-- KPI Section -->

<section class="kpi-section">

<div class="container">

<div class="row g-4">

<div class="col-lg-3 col-md-6">

<div class="kpi-card">

<h2>95%</h2>

<p>Charging Efficiency</p>

</div>

</div>

<div class="col-lg-3 col-md-6">

<div class="kpi-card">

<h2>620 KM</h2>

<p>Maximum Range</p>

</div>

</div>

<div class="col-lg-3 col-md-6">

<div class="kpi-card">

Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

<h2>75 kWh</h2>

<p>Battery Capacity</p>

</div>

</div>

<div class="col-lg-3 col-md-6">

<div class="kpi-card">

<h2>45 Min</h2>

<p>Fast Charging Time</p>

</div>

</div>

</div>

</div>

</section>

{% endblock %}

about.html

{% extends 'base.html' %}

{% block content %}

<div class="container py-5">

<h1 class="section-title text-center">

Project Team

</h1>

<div class="row g-4">

<div class="col-md-4">

<div class="card-custom text-center">

Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

<h3>CHINTHAPARTHI VANAJA</h3>

<p>Team Leader</p>

<p>SkillWallet ID: SWUID2026202797</p>

</div>

</div>

<div class="col-md-4">

<div class="card-custom text-center">

<h3>TUPAKULA SRAVANTHI</h3>

<p>TEAM MEMBER</p>

<p>SkillWallet ID: SWUID2026202814</p>

</div>

</div>

<div class="col-md-4">

<div class="card-custom text-center">

<h3>KOKKANTI HARITHA</h3>

<p>TEAM MEMBER</p>

<p>SkillWallet ID: SWUID2026202801</p>

</div>

</div>

</div>

</div>

{% endblock %}

dashboard01.html

{% extends 'base.html' %}

{% block content %}

<div class="container py-5">

Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

```
<h1 class="section-title">
  Charging Dashboard
</h1>

<div class="chart-card mb-4">
  <h4>Charging Analysis</h4>
  <p>Interactive Tableau dashboard showing charging analytics.</p>
</div>

<div class="tableau-container">
  <div class="tableauPlaceholder" id="tableauViz">
    <noscript>
      <a href="#">
        
      </a>
    </noscript>

    <object class="tableauViz" style="display:none;">
      <param name="host_url" value="https://public.tableau.com/" />
      <param name="embed_code_version" value="3" />
      <param name="path" value="shared/HY8BXGGZP" />
      <param name="toolbar" value="yes" />
      <param name="animate_transition" value="yes" />
      <param name="display_spinner" value="yes" />
      <param name="display_overlay" value="yes" />
      <param name="display_count" value="yes" />
      <param name="language" value="en-GB" />
    </object>
  </div>
</div>

<div>

</div>

<style>
.tableau-container {
  width: 100%;
  overflow-x: auto;
}
```

Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

```
.tableauPlaceholder {
    width: 100%;
    min-height: 850px;
}

.chart-card {
    background: #fff;
    padding: 20px;
    border-radius: 12px;
    box-shadow: 0 2px 8px rgba(0,0,0,0.1);
    margin-bottom: 20px;
}

.section-title {
    margin-bottom: 20px;
}

</style>

<script>
document.addEventListener("DOMContentLoaded", function () {
    var divElement = document.getElementById('tableauViz');
    var vizElement = divElement.getElementsByTagName('object')[0];

    vizElement.style.width = '100%';
    vizElement.style.height = '850px';

    var scriptElement = document.createElement('script');
    scriptElement.src =
        'https://public.tableau.com/javascripts/api/viz_v1.js';
    vizElement.parentNode.insertBefore(scriptElement, vizElement);
});
</script>

{% endblock %}
```

Dashboard02

```
{% extends 'base.html' %}
```

```
{% block content %}
```

```
<div class="container py-5">
```

```
<h1 class="section-title">
```

```
    Range Dashboard
```

```
</h1>
```

Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

```
<div class="chart-card mb-4">
  <h4>Range Analysis</h4>
  <p>Vehicle range charts can be displayed here.</p>
</div>

<div class="tableau-container">
  <div class="tableauPlaceholder" id="tableauViz">
    <noscript>
      <a href="#">
        
      </a>
    </noscript>

    <object class="tableauViz" style="display:none;">
      <param name="host_url" value="https://public.tableau.com/" />
      <param name="embed_code_version" value="3" />
      <param name="site_root" value="" />
      <param name="name"
value="visualizationToolforElectricVehicleChargeandRangeAnalysisusingTable
  au-VANAJA-DASHBOARD02/Dashboard2" />
      <param name="tabs" value="no" />
      <param name="toolbar" value="yes" />
      <param name="animate_transition" value="yes" />
      <param name="display_spinner" value="yes" />
      <param name="display_overlay" value="yes" />
      <param name="display_count" value="yes" />
      <param name="language" value="en-GB" />
    </object>
  </div>
</div>

<style>
.tableau-container {
  width: 100%;
  overflow-x: auto;
}
```

Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

```
.tableauPlaceholder {
    width: 100%;
    min-height: 850px;
}

.chart-card {
    background: #fff;
    padding: 20px;
    border-radius: 12px;
    box-shadow: 0 2px 8px rgba(0,0,0,0.1);
    margin-bottom: 20px;
}

.section-title {
    margin-bottom: 20px;
}

</style>

<script>
document.addEventListener("DOMContentLoaded", function () {
    var divElement = document.getElementById('tableauViz');
    var vizElement = divElement.getElementsByTagName('object')[0];

    vizElement.style.width = '100%';
    vizElement.style.height = '850px';

    var scriptElement = document.createElement('script');
    scriptElement.src =
        'https://public.tableau.com/javascripts/api/viz_v1.js';
    vizElement.parentNode.insertBefore(scriptElement, vizElement);
});
</script>

{% endblock %}
```

Dashboardjs-01

```
new Chart(document.getElementById('chargeChart'), {
    type: 'line',

    data: {
        labels: ['0', '20', '40', '60', '80', '100'],

        datasets: [{
            label: 'Charging Time',
```

Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

```
data:[0,12,28,48,80,120],  
borderColor:'#60A5FA'  
    }]  
    }  
});
```

```
new Chart(document.getElementById('batteryChart'),{  
    type:'bar',  
    data:{  
labels:['Tesla','BYD','MG','Nexon'],  
        datasets:[{  
label:'Battery Capacity',  
            data:[75,60,50,40]  
        }]  
    }  
});
```

Dashboardjs-02

```
new Chart(document.getElementById('rangeChart'),{  
    type:'bar',  
    data:{  
labels:['Tesla','BYD','MG','Nexon'],  
        datasets:[{  
label:'Range (km)',  
            data:[620,520,430,465]  
        }]  
    }  
});
```

```
new Chart(document.getElementById('effChart'),{  
    type:'doughnut',  
    data:{  
labels:['Battery','Motor','Cooling','Others'],  
        datasets:[{  
            data:[45,30,15,10]  
        }]  
    }  
});
```

Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

```
}  
});
```

Story

```
{% extends 'base.html' %}  
  
{% block content %}  
  
<div class="container py-5">  
  
<h1 class="section-title">  
    EV Storytelling  
</h1>  
  
<div class="chart-card mb-4">  
<h4>Electric Vehicle Charge and Range Story</h4>  
<p>  
Interactive storytelling dashboard showing EV charging patterns,  
battery performance, and vehicle range insights.  
</p>  
</div>  
  
<div class="tableau-container">  
<div class="tableauPlaceholder" id="tableauViz">  
    <noscript>  
        <a href="#">  
              
            </a>  
        </noscript>  
  
<object class="tableauViz" style="display:none;">  
<param name="host_url" value="https://public.tableau.com/" />  
<param name="embed_code_version" value="3" />  
<param name="path" value="shared/ZGB2HFKH8" />  
<param name="toolbar" value="yes" />  
<param name="animate_transition" value="yes" />  
<param name="display_spinner" value="yes" />  
<param name="display_overlay" value="yes" />  
<param name="display_count" value="yes" />  
<param name="language" value="en-GB" />
```

```
</object>
</div>
</div>

</div>

<style>
.tableau-container {
  width: 100%;
  overflow-x: auto;
}

.tableauPlaceholder {
  width: 100%;
  min-height: 1000px;
}

.chart-card {
  background: #fff;
  padding: 20px;
  border-radius: 12px;
  box-shadow: 0 2px 8px rgba(0,0,0,0.1);
  margin-bottom: 20px;
}

.section-title {
  margin-bottom: 20px;
}
</style>

<script>
document.addEventListener("DOMContentLoaded", function () {
  var divElement = document.getElementById('tableauViz');
  var vizElement = divElement.getElementsByTagName('object')[0];

  vizElement.style.width = '100%';
  vizElement.style.height = '1000px';

  var scriptElement = document.createElement('script');
  scriptElement.src =
    'https://public.tableau.com/javascripts/api/viz_v1.js';
  vizElement.parentNode.insertBefore(scriptElement, vizElement);
});
</script>
```

Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

```
{% endblock %}
```

style.css

```
:root{
  --bg:#050816;
  --card:#0F172A;
  --blue:#2563EB;
  --light-blue:#60A5FA;
  --text:#F8FAFC;
}

body{
  background:var(--bg);
  color:var(--text);
  font-family:Arial,sans-serif;
}

.navbar{
  background:#000814 !important;
}

.navbar-brand{
  color:var(--light-blue)!important;
  font-weight:bold;
}

.hero{
  min-height:80vh;
  display:flex;
  justify-content:center;
  align-items:center;
  text-align:center;
}

.card-custom{
  background:var(--card);
  border-radius:15px;
  padding:25px;
  border:1px solid #1E293B;
  height:100%;
}

.section-title{
  color:var(--light-blue);
  margin-bottom:30px;
}
```


Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

```
font-weight:700;
}

.btn-ev{
background:var(--blue);
color:white;
border:none;
padding:10px 25px;
border-radius:25px;
transition:0.3s;
}

.btn-ev:hover{
background:var(--light-blue);
color:#000;
}

.chart-card{
background:var(--card);
padding:20px;
border-radius:15px;
border:1px solid #1E293B;
margin-bottom:25px;
}

.tableau-container{
width:100%;
overflow-x:auto;
background:var(--card);
border-radius:15px;
padding:15px;
border:1px solid #1E293B;
}

.tableauPlaceholder{
width:100%;
min-height:850px;
}

.container{
max-width:1400px;
}

footer{
background:#000814;
```

Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

```
color:var(--text);  
text-align:center;  
padding:15px;  
margin-top:40px;  
}
```

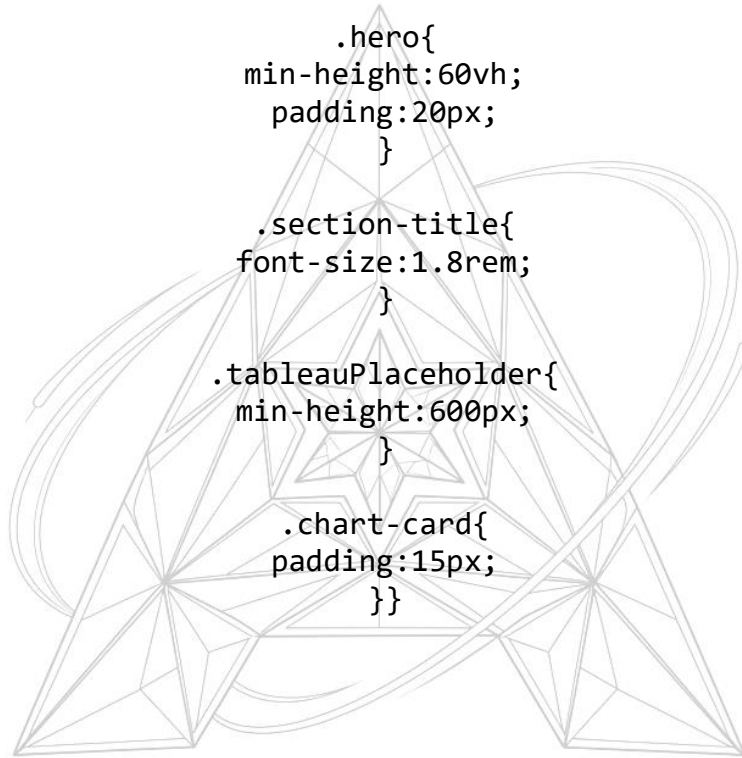
```
@media (max-width:768px){
```

```
  .hero{  
    min-height:60vh;  
    padding:20px;  
  }
```

```
  .section-title{  
    font-size:1.8rem;  
  }
```

```
  .tableauPlaceholder{  
    min-height:600px;  
  }
```

```
  .chart-card{  
    padding:15px;  
  }}
```



ALPHA
— INNOVATORS —

